Zhiwei Li

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EDUCATION

Duke University & Duke Kunshan University Dual Degree (Class of 2028)
BS: Interdisciplinary Studies (Subplans: Applied Math & Computational Science; Computer Science), Duke University; Applied Math & Computational Science; Computer Science (Duke Kunshan University)

- GPA: 3.85/4.00; Dean's List Fall 2024, Dean's List Spring 2025, Dean's List Fall 2025
- Courses: Machine Learning, Programming & Database, Probability & Statistics, Computer Graphics

INTERNSHIP

HP Inc., Beijing, Data Analyst Intern 12/2025-02/2026

- Designed **data processing pipelines** for supply chain analytics using Python and Pandas, transforming fragmented operational data into structured analytical datasets. Built **modular automation scripts** for data ingestion, validation, transformation, and report generation, significantly reducing manual processing effort.
- Collaborated with operations managers to model **end-to-end logistics workflows** and design structured reporting systems for supply chain monitoring. Developed reusable automation tools currently **deployed internally** to support ongoing operational analytics.

PROJECT EXPERIENCE

ChatDKU — LLM Agent System with RAG, Core Research Member 08/2025-Present
Supervisor: Prof. Bing Luo, Assistant Professor of Data and Computational Science Kunshan, China

- Built **production-scale RAG pipelines** using **DSPy and LlamaIndex**, integrating structured and unstructured institutional knowledge sources. Developed **automated data ingestion pipelines**, including web crawling, content parsing, schema normalization, and scheduled updates to maintain a continuously evolving knowledge base.
- Engineered backend infrastructure using **PostgreSQL, Redis, and Chroma vector databases** to support high-performance retrieval and multi-user workloads. Deployed and maintained **lab-scale AI services and GPU servers**, performing performance monitoring, concurrency debugging, and system optimization.
- Collaborated with multiple university offices to deploy customized AI assistants for different administrative departments with support from the **Office of the Chancellor**.

Motion-Based Bird Species Classification from Surveillance Video, Project Team Lead 11/2025-03/2026
Supervisor: Prof. Kaizhu Huang, Professor of Electrical and Computer Engineering Kunshan, China

- Integrated **YOLOv8 detection, DeepLabCut pose estimation, and pose tracking** to extract skeletal motion trajectories. Built temporal motion including **joint articulation metrics, and frequency-domain behavioral signals**.
- Implemented model training pipelines in **PyTorch**, enabling efficient GPU experimentation and model iteration. Constructed classification models using **sparse feature selection and statistical learning** to identify motion signatures for species discrimination.

NaviRoom — AI-Powered Intelligent Room Recommendation System, Project Team Lead 08/2025-Present
Supervisor: Prof. Dongmian Zou, Assistant Professor of Data Science Kunshan, China

- Developed an **AI-driven platform for automatic reservation system generation** from heterogeneous inputs including CSV, Excel, and natural language descriptions. Built **LLM-based NLP pipelines** using **DSPy and BERT models** to perform schema recognition and semantic extraction from unstructured room descriptions.
- Designed a **zero-shot recommendation engine** combining semantic similarity and content-based filtering to address the cold-start problem in scheduling systems. Implemented a **multi-tenant SaaS architecture** that automatically generates organization-specific databases and APIs while maintaining a unified codebase.

RESEARCH EXPERIENCE

Global Study on the Future of AI Education, Research Team Lead 09/2024-07/2025
U-Corp Program at DKU (Enterprise Collaborator: Eric Liu, CEO of Philosophie AI) Kunshan, China

- Conducted a **large-scale literature review** on institutional AI in higher education. Designed survey instruments and conducted interviews with educators on AI integration in teaching and research.
- Led a cross-disciplinary research team and presented findings to **100+ participants** in institutional seminars.

ADDITIONAL INFORMATION

Languages: Chinese Mandarin (Native), English (Full Professional Proficiency, GRE 330)
Prize: International Genetically Engineered Machine Competition (iGEM) – Silver Medal
Programming Language： Java, Python, SQL (MySQL，PostgresSQL), HTML/CSS/JS
Dev Tools： Git / GitHub, Linux / macOS Dev Environment
Design & Digital Media Tools: Photoshop, Processing, Premiere, Illustrator, After Effects